

# Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

## 1. Цели и задачи работы

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Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

## 2. Процесс выполнения лабораторной работы

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# Работа с Midnight Commander

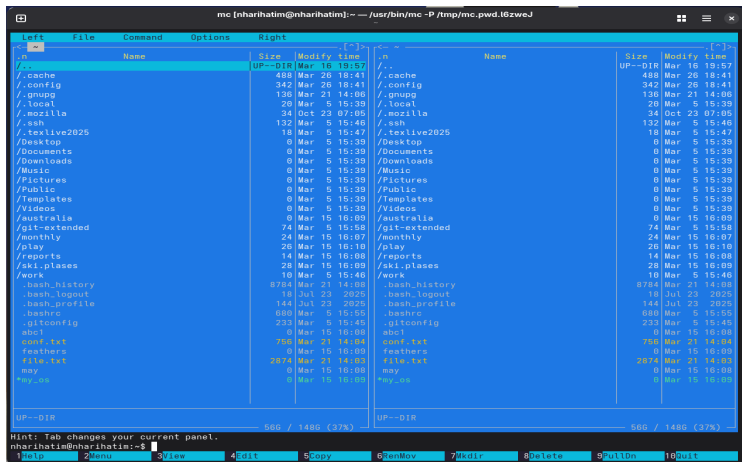


Рисунок 1: Запуск mc

# Работа с Midnight Commander

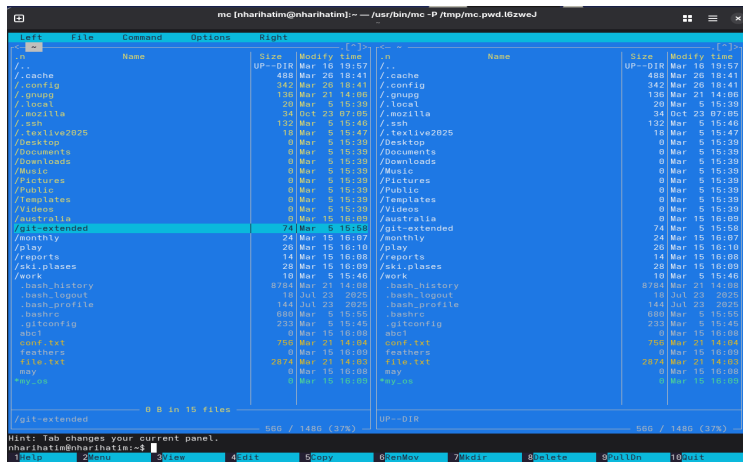


Рисунок 2: Выделение



# Работа с Midnight Commander

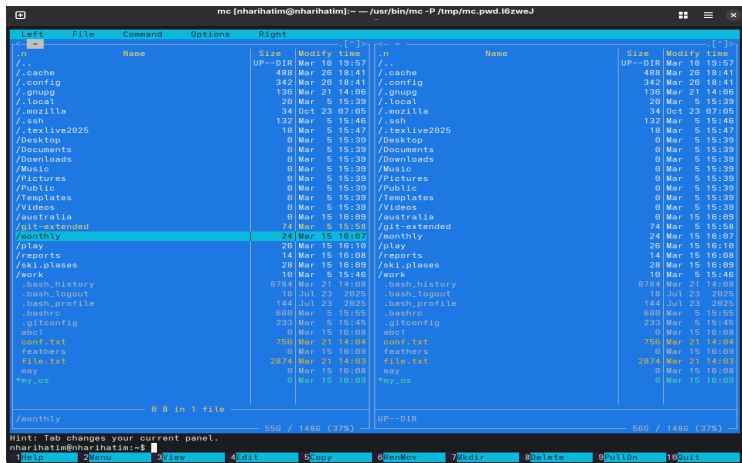


Рисунок 3: Отмена

# Работа с Midnight Commander

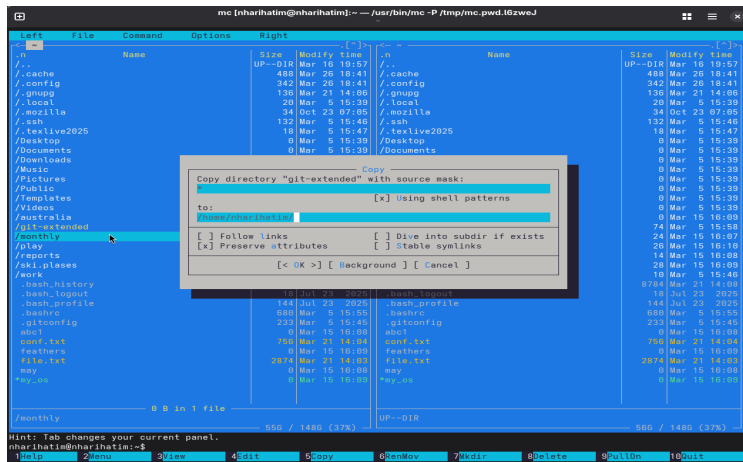


Рисунок 4: Копирование

# Работа с Midnight Commander

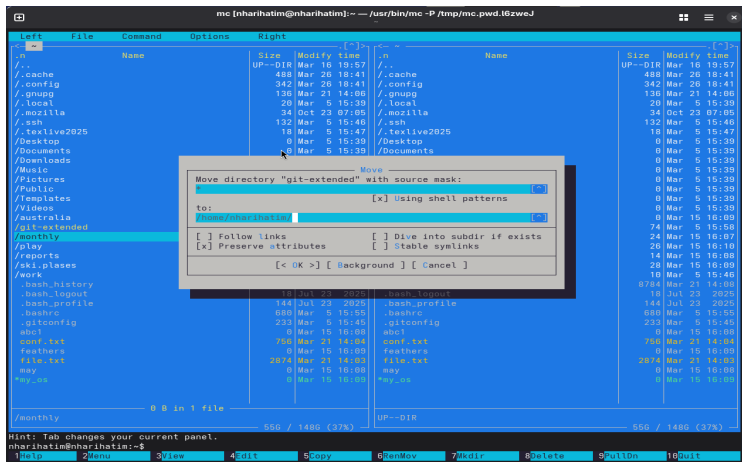


Рисунок 5: Перемещение

# Работа с Midnight Commander

The screenshot shows the Midnight Commander (MC) interface. The top bar displays the command prompt: `mc [nharihatim@nharihatim]:~ - /usr/bin/mc -P /tmp/mc.pwd.l6zweJ`. The main window is divided into two panes. The left pane shows a directory listing with columns for Name, Size, and Modify time. The right pane shows detailed information for the selected file, `/monthly`.

| Left                       | File                 | Command                   | Options | Right                                                   |
|----------------------------|----------------------|---------------------------|---------|---------------------------------------------------------|
| <code>&lt;-</code>         | <code>..</code>      |                           |         | <code>Size</code> <code>Modify</code> <code>time</code> |
| <code>..</code>            | <code>UP--DIR</code> |                           |         | <code>Mar 15 19:57</code>                               |
| <code>./cache</code>       | <code>488</code>     | <code>Mar 26 18:41</code> |         |                                                         |
| <code>./config</code>      | <code>342</code>     | <code>Mar 26 18:41</code> |         |                                                         |
| <code>./gnupg</code>       | <code>138</code>     | <code>Mar 21 14:08</code> |         |                                                         |
| <code>./local</code>       | <code>29</code>      | <code>Mar 5 15:39</code>  |         |                                                         |
| <code>./mozilla</code>     | <code>34</code>      | <code>Oct 23 07:05</code> |         |                                                         |
| <code>./ssh</code>         | <code>132</code>     | <code>Mar 5 15:46</code>  |         |                                                         |
| <code>./texlive2025</code> | <code>18</code>      | <code>Mar 5 15:47</code>  |         |                                                         |
| <code>/Desktop</code>      | <code>0</code>       | <code>Mar 5 15:39</code>  |         |                                                         |
| <code>/Documents</code>    | <code>0</code>       | <code>Mar 5 15:39</code>  |         |                                                         |
| <code>/Downloads</code>    | <code>0</code>       | <code>Mar 5 15:39</code>  |         |                                                         |
| <code>/Music</code>        | <code>0</code>       | <code>Mar 5 15:39</code>  |         |                                                         |
| <code>/Pictures</code>     | <code>0</code>       | <code>Mar 5 15:39</code>  |         |                                                         |
| <code>/Public</code>       | <code>0</code>       | <code>Mar 5 15:39</code>  |         |                                                         |
| <code>/Templates</code>    | <code>0</code>       | <code>Mar 5 15:39</code>  |         |                                                         |
| <code>/Videos</code>       | <code>0</code>       | <code>Mar 5 15:39</code>  |         |                                                         |
| <code>/australia</code>    | <code>0</code>       | <code>Mar 15 16:09</code> |         |                                                         |
| <code>/git-extended</code> | <code>74</code>      | <code>Mar 5 15:58</code>  |         |                                                         |
| <code>/monthly</code>      | <code>24</code>      | <code>Mar 15 16:07</code> |         |                                                         |
| <code>/play</code>         | <code>26</code>      | <code>Mar 15 16:10</code> |         |                                                         |
| <code>/reports</code>      | <code>14</code>      | <code>Mar 15 16:08</code> |         |                                                         |
| <code>/ski.places</code>   | <code>28</code>      | <code>Mar 15 16:09</code> |         |                                                         |
| <code>/work</code>         | <code>18</code>      | <code>Mar 5 15:46</code>  |         |                                                         |
| <code>.bash_history</code> | <code>8784</code>    | <code>Mar 21 14:08</code> |         |                                                         |
| <code>.bash_logout</code>  | <code>18</code>      | <code>Jul 23 2025</code>  |         |                                                         |
| <code>.bash_profile</code> | <code>144</code>     | <code>Jul 23 2025</code>  |         |                                                         |
| <code>.bashrc</code>       | <code>680</code>     | <code>Mar 5 15:55</code>  |         |                                                         |
| <code>.gitconfig</code>    | <code>233</code>     | <code>Mar 5 15:46</code>  |         |                                                         |
| <code>abc1</code>          | <code>0</code>       | <code>Mar 15 16:08</code> |         |                                                         |
| <code>conf.txt</code>      | <code>750</code>     | <code>Mar 21 14:04</code> |         |                                                         |
| <code>feathers</code>      | <code>0</code>       | <code>Mar 15 16:09</code> |         |                                                         |
| <code>file.txt</code>      | <code>2874</code>    | <code>Mar 21 14:02</code> |         |                                                         |
| <code>may</code>           | <code>0</code>       | <code>Mar 15 16:08</code> |         |                                                         |
| <code>*my.os</code>        | <code>0</code>       | <code>Mar 15 16:08</code> |         |                                                         |

Information for `/monthly`:

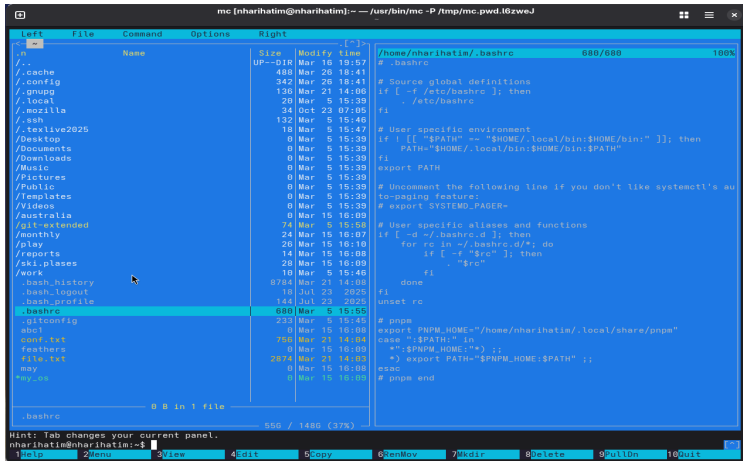
- File: `monthly`
- Location: `2Bh:3DEB99h`
- Mode: `drwx--x--x (0711)`
- Attributes: `-----`
- Links: `1`
- Owner: `nharihatim/nharihatim`
- Size: `24 (0 blocks)`
- Changed: `Mar 15 16:08`
- Modified: `Mar 15 16:07`
- Accessed: `Mar 26 18:40`
- Filesystem: `/home`
- Device: `/dev/nvme0n1p3`
- Type: `ntfs (21h)`
- Free space: `556 / 1486 (37%)`
- No node information

Bottom status bar: `Hint: Tab changes your current panel. nharihatim@nharihatim:~$`

Footer: `1|Help 2|Menu 3|View 4|Edit 5|Copy 6|RenMov 7|Mkdir 8|Delete 9|PullDn 10|Quit`

Рисунок 6: Информация

## Работа с Midnight Commander



### Рисунок 7: Быстрый просмотр

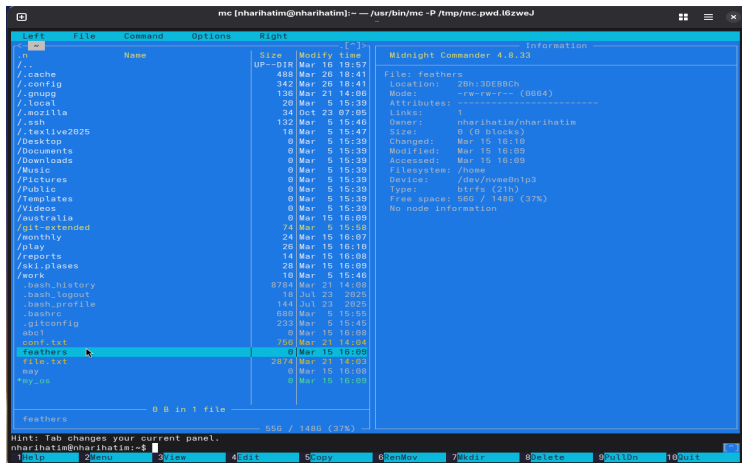


Рисунок 8: Информация

# Работа с Midnight Commander

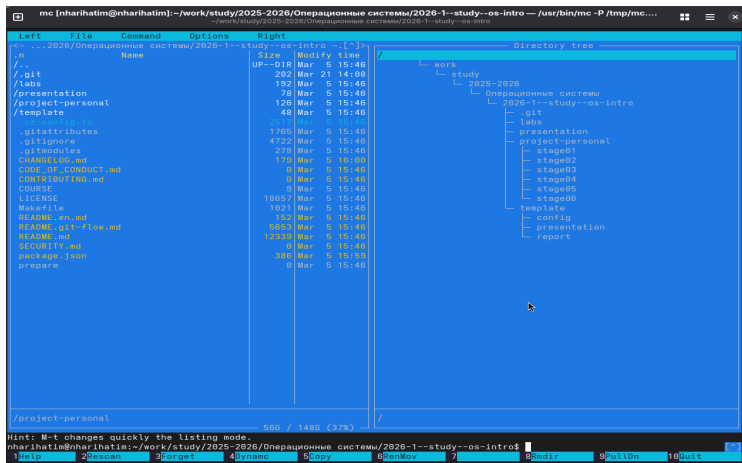
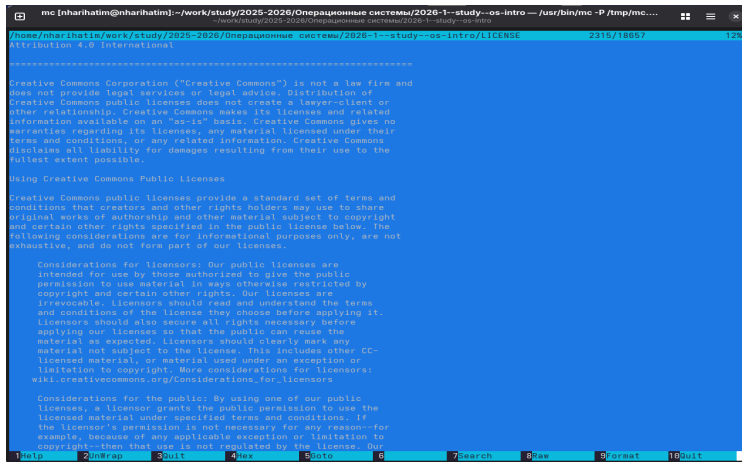


Рисунок 9: Дерево каталогов

# Работа с Midnight Commander



The screenshot shows the Midnight Commander (mc) interface. The title bar indicates the user is 'nharihatim' and the current directory is '/work/study/2025-2026/Операционные системы/2026-1--study--os-intro'. The main window displays the contents of a file named 'LICENSE' in the directory '/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/'. The file size is 2315/18657 bytes. The license text is displayed in a monospaced font on a blue background. At the bottom, there is a status bar with keyboard shortcuts: 1)help, 2)Unmap, 3)Quit, 4)Hex, 5)Spots, 6), 7)Search, 8)Zw, 9)Format, 10)Quit.

```
mc [nharihatim@nharihatim]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro — /usr/bin/mc -P /tmp/mc....
~work/study/2025-2026/Операционные системы/2026-1--study--os-intro

/home/nharihatim/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/LICENSE 2315/18657 12%
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```

Рисунок 10: Просмотр содержимого текстового файла



# Работа с Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the user is 'mc [nharihatim@nharihatim]' in the directory '~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro'. The main window displays the text of the 'Attribution 4.0 International' license. The text is as follows:

```
License [-----] 0 L: [ 1+ 0 1/397] *(0 /18657b) 0065 0x041 [*](X)
Attribution 4.0 International

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```

At the bottom of the window, there is a status bar with the following shortcuts: 1 Help, 2 Save, 3 Mark, 4 Replace, 5 Copy, 6 Move, 7 Search, 8 Delete, 9 Pull Down, 10 Quit.

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

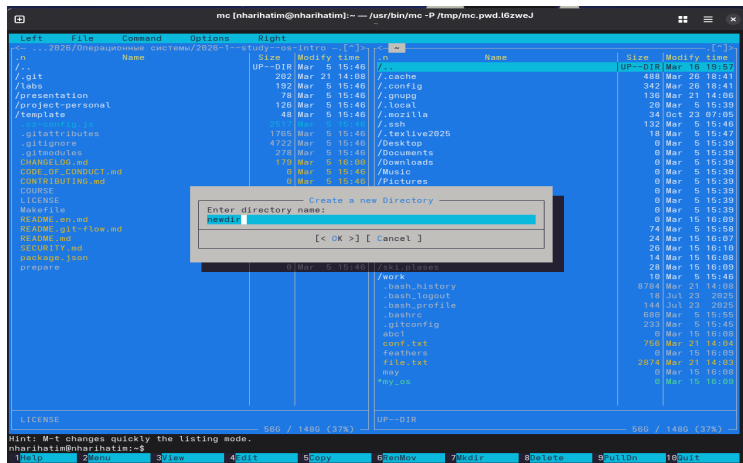
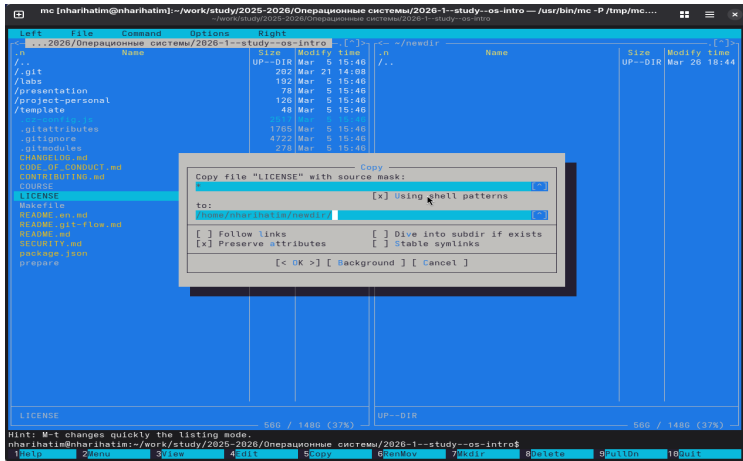


Рисунок 12: Создание каталога

## Работа с Midnight Commander



**Рисунок 13:** Копирование в файлов в созданный каталог

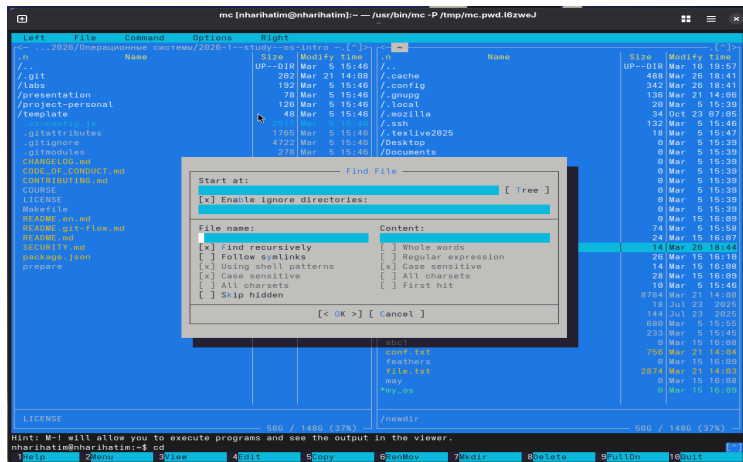


Рисунок 14: Поиск файлов

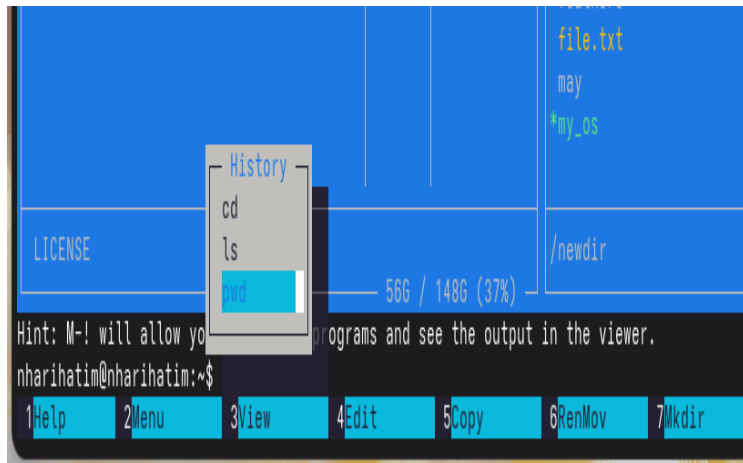


Рисунок 15: История команд

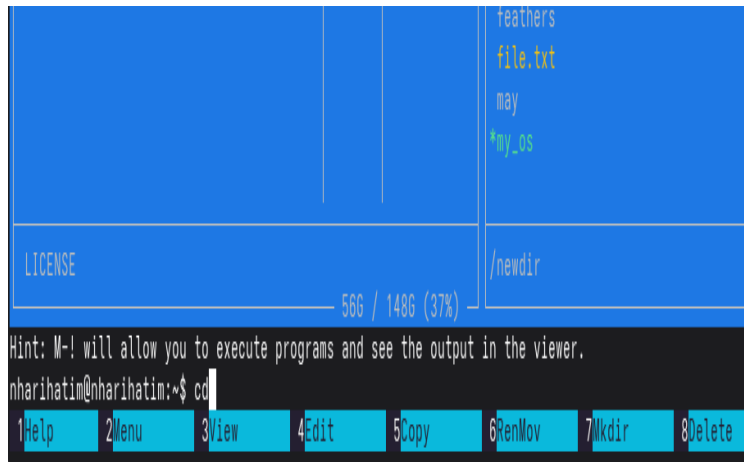
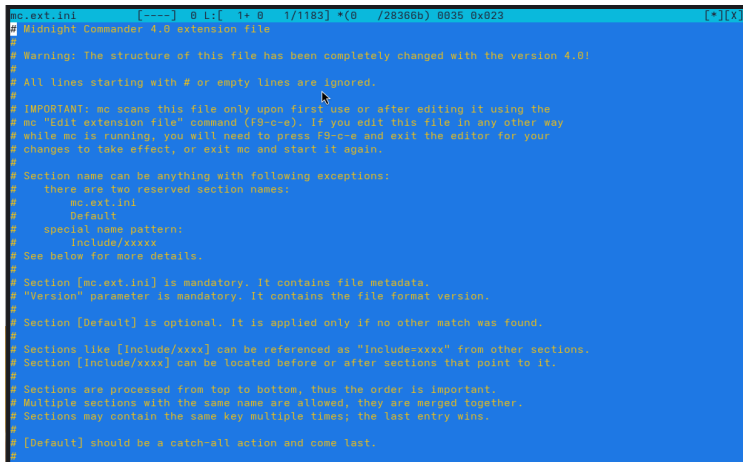


Рисунок 16: Переход в домашний каталог



```
mc.ext.ini      [---] 0 L: 1+ 0 1/1183 *(0 /28366b) 0035 0x023      [*][X]
# Midnight Commander 4.8 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.8!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [Include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
```

Рисунок 17: Просмотр файла расширений

```
menu [----] 0 L:[ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [X]
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#           %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
#   ascii, hex, proff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
#   %f   quote expanded macro
#   %if  ditto
#   %0f  don't quote expanded macro
#####

+ ! t t
0   Do something on the current file
    CMD=%{Enter command}
    $CMD %f

+ t t
0   Do something on the tagged files
    CMD=%{Enter command}
    for i in %t ; do
```

Рисунок 18: Просмотр файла меню



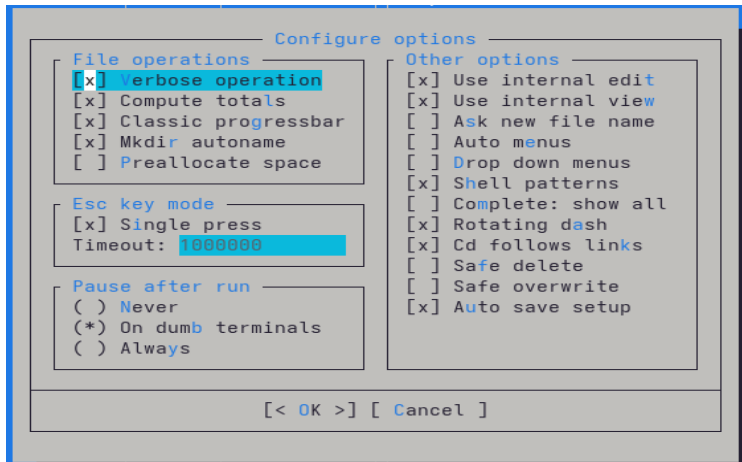


Рисунок 19: Конфигурация

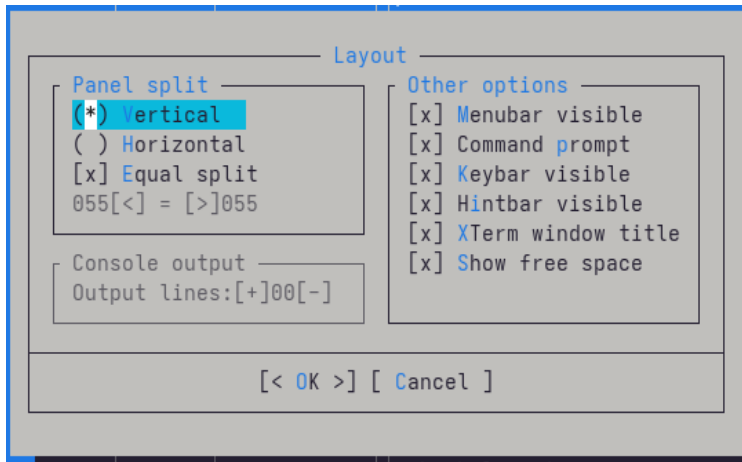


Рисунок 20: Внешний вид

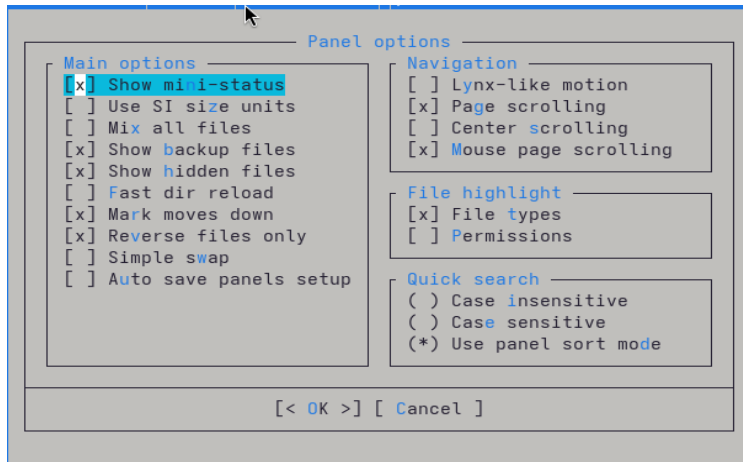


Рисунок 21: Настройки панелей

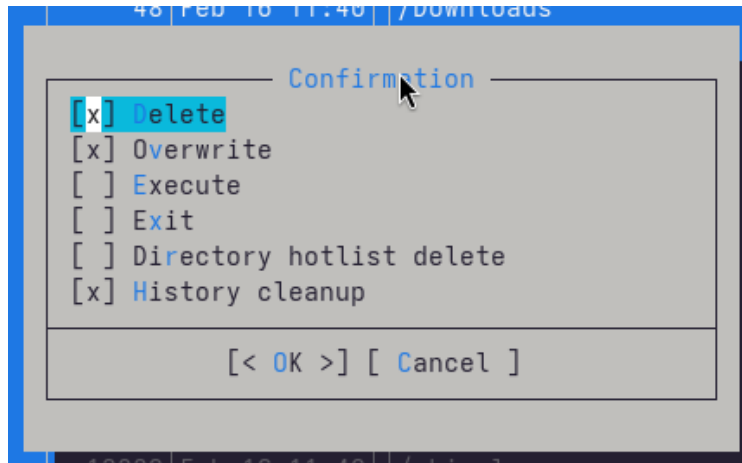


Рисунок 22: Подтверждение

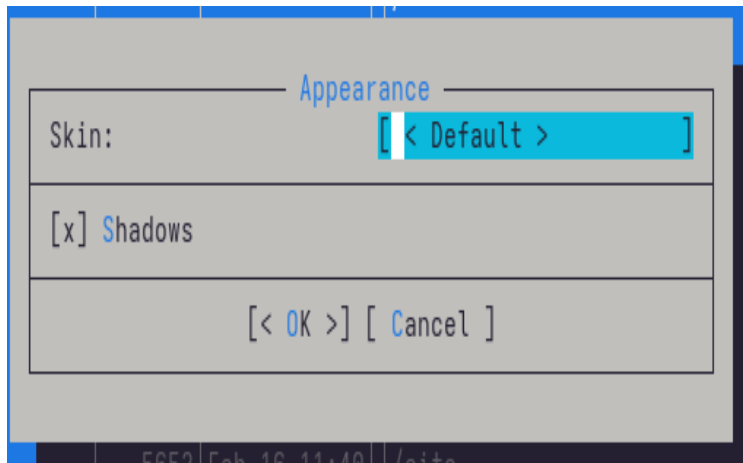


Рисунок 23: Оформление

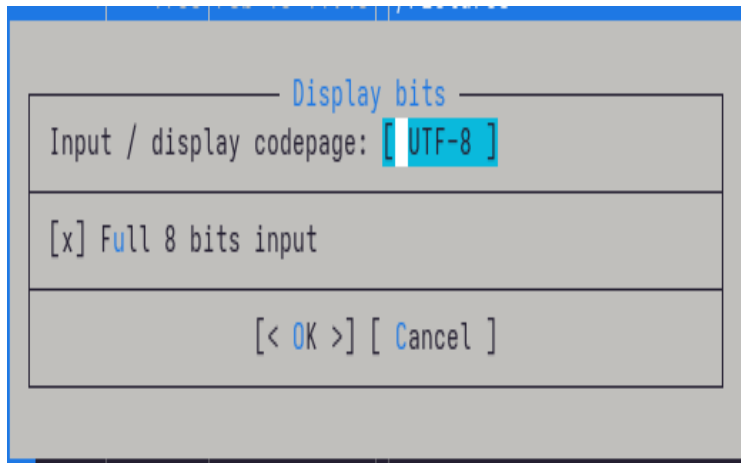


Рисунок 24: Кодировка символов

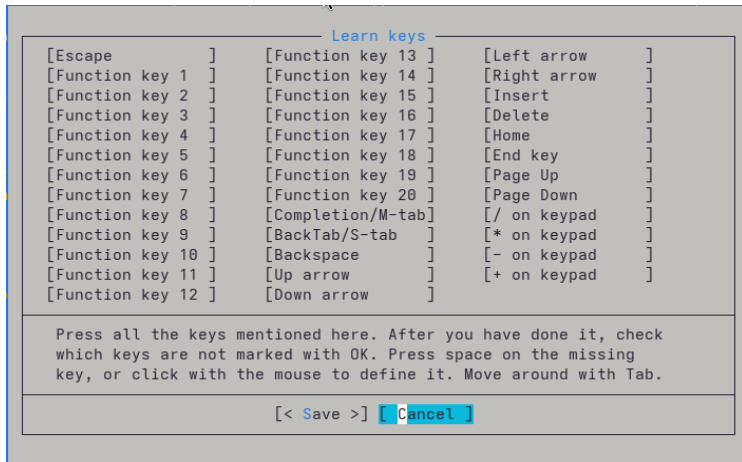
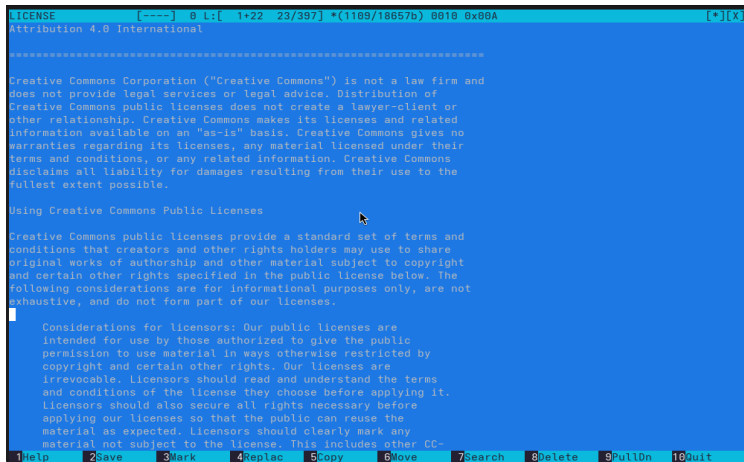


Рисунок 25: Распознавание клавиш



```
LICENSE      [-----]  8 L:[ 1+22 23/397] *(1109/18657b) 0010 0x00A      [*][X]
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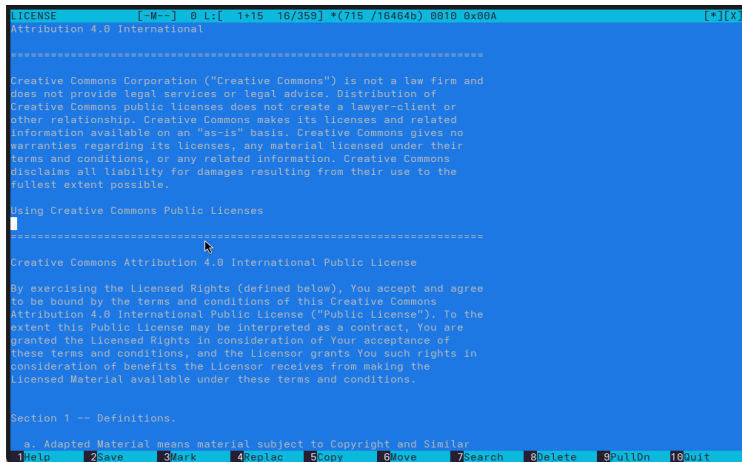
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```

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit

Рисунок 26: Файл с текстом





```
LICENSE      [-M--]  8 L:[ 1+15 16/359] *(715 /16464b) 0010 0x00A      [*][x]
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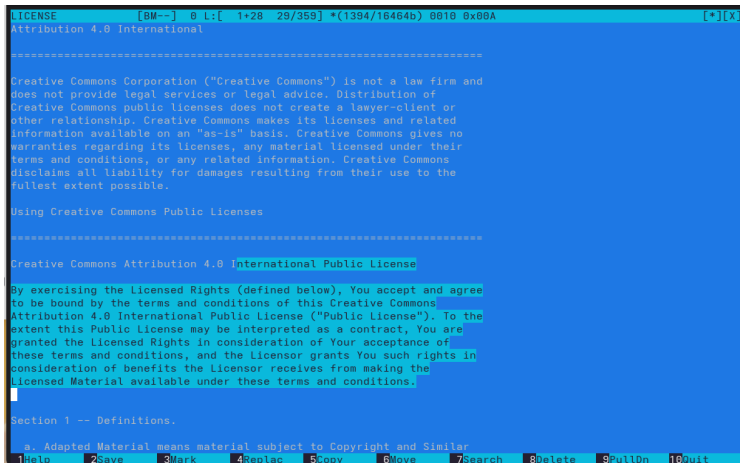
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Section 1 -- Definitions.

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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 27: Файл с текстом



```
LICENSE      [BM--]  0 L:[ 1+28 29/359] *(1394/16464b) 0010 0x00A      [*][X]
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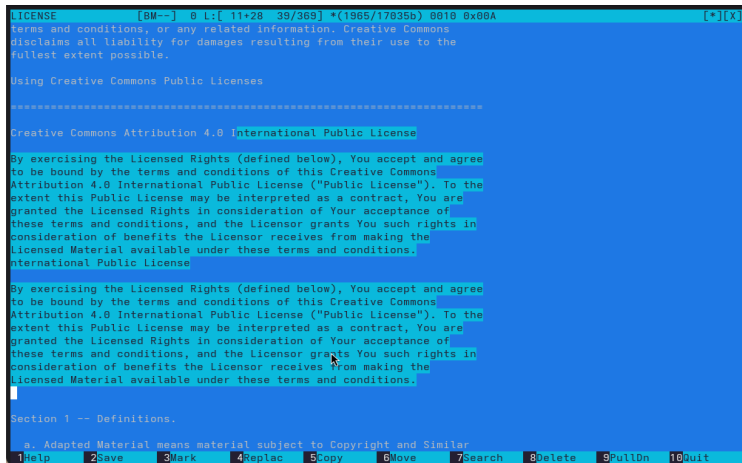
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```

Рисунок 28: Копирование фрагмента



```
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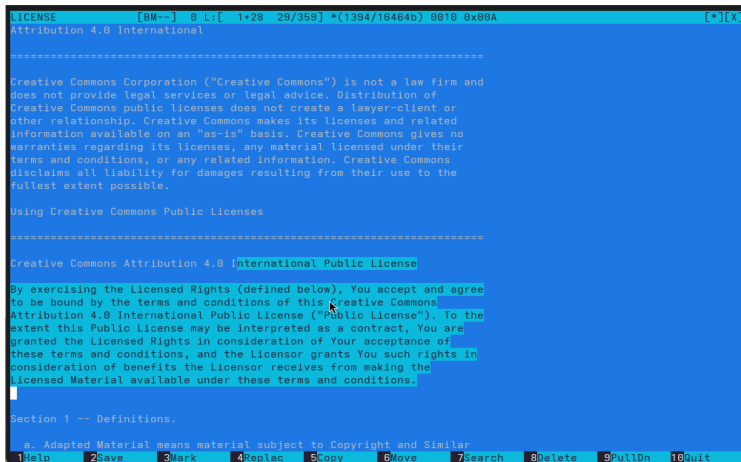
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```

Рисунок 29: Сохранение



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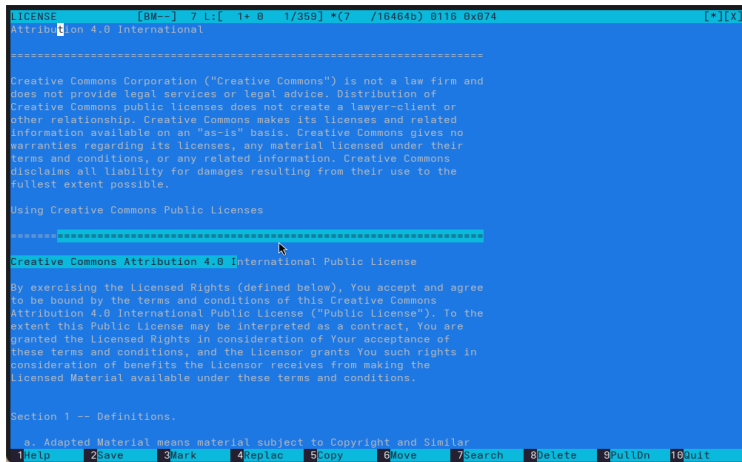
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```

Рисунок 30: Отмена



Рисунок 31: Переход в конец файла



```
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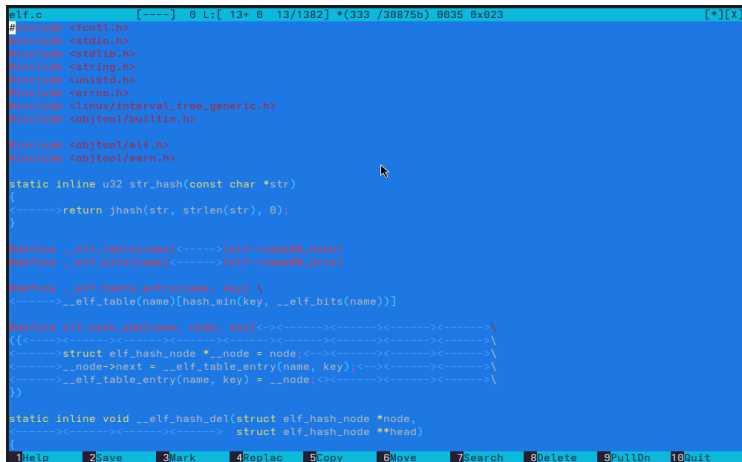
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```

Рисунок 32: Переход в начало файла



```
elf.c [----] 0 L: [ 13+ 0 13/1382] *(333 /30875b) 0035 0x023 [*][X]
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <errno.h>
#include <linux/interval_tree_generic.h>
#include <objtool/builtin.h>

#include <objtool/elf.h>
#include <objtool/warn.h>

static inline u32 str_hash(const char *str)
{
    ----->return jhash(str, strlen(str), 0);
}

#define __elf_table(name)----->{elf->name##_hash}
#define __elf_bits(name)----->{elf->name##_bits}

#define __elf_table_entry(name, key) \
    ----->__elf_table(name)[hash_min(key, __elf_bits(name))]

#define elf_hash_add(name, node, key)----->----->----->\
    ({----->----->----->----->----->\
    ----->struct elf_hash_node *__node = node;----->----->----->\
    ----->__node->next = __elf_table_entry(name, key);----->----->----->\
    ----->__elf_table_entry(name, key) = __node;----->----->----->\
    })

static inline void __elf_hash_del(struct elf_hash_node *node,
    ----->----->-----> struct elf_hash_node **head)
```

1Help 2Save 3/ark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit

Рисунок 33: Файл с программой

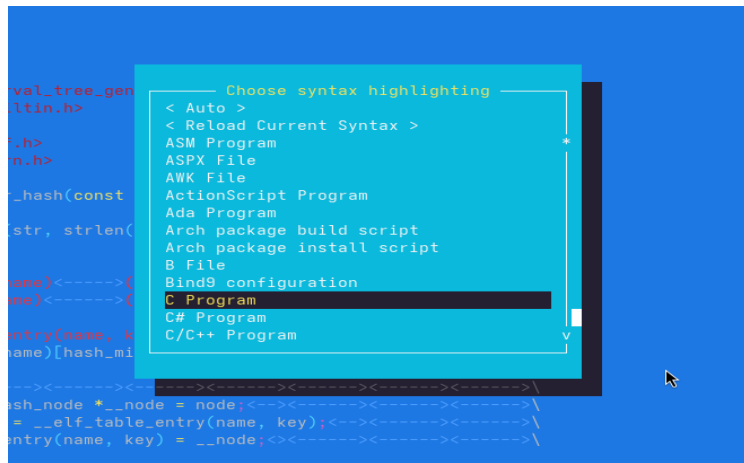


Рисунок 34: Цветовыделение синтаксиса



### 3. Выводы по проделанной работе

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В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.